Power BI Assignment 1 –Data Transformation & Data Modeling

E-Commerce Sales Analysis

This assignment will help you explore e-commerce sales data analysis using Power BI.
Below are the files you will be working with (click on each to download):

★ List of Orders.csv
★ Order Details.csv
★ Sales target.csv

In this exercise, you will leverage Power BI's capabilities to import, transform, model, and analyze the provided data.

Instructions:-

Import Data:

● Import "List of Orders.csv" into Power BI.
● Open "List of Orders" in Power Query Editor by clicking on 'Transform'.
   Get Data→Text/CSV→Select "List of Orders.csv" →open→Transform
● Import "Order Details.csv" and "Sales target.csv" into Power Query Editor.
Data Transformation:
   Get Data→Text/CSV→Select "Order Details.csv" →open→Transform
   Get Data→Text/CSV→Select "Sales target.csv" →open→Transform

● Restrict the "List of Orders" table to only the first 500 rows.
   Select list of orders – Keep rows- keep top rows – number of rows – 500 ok

● Ensure the "Order Date" column in the "List of Orders" table is set to data type 'Date'.
   Select order date→Right click→change type→Date→change date→but some fields occurs "Error" So
   Change type→use locale→Date→English(United Kingdom)→ok

● Change the data type of "Amount" and "Target" columns to 'Fixed Decimal Number'.
   Order details→select amount→change type→ "Fixed Decimal Number"

● Format the "CustomerName" column into proper case, ensuring consistent capitalization for each word.
   Change type→Capitalize for Each Word

● Merge the "State" and "City" columns to create a new column named "Location" in the format 'City, State'.
   Add new column→Location→Insert = [CITY] & " " & [STATE]

● Create a new custom column named "Profit Margin" as the percentage of "Profit" divided by "Amount".

Add new column→Profit Margin→Profit/Amount→ok→ Change type→Fixed Decimal Number

● Add a new conditional column named "Profit Status" based on the values in the "Profit" column. The conditions are as follows: if the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

Add condition column-

New Column name→Profit Status

Column name	Operator	Value	Output
Profit	is less than	0	Loss
Profit	equals	0	Break-Even

Else

Profit

Merging Data (Joins):

● Merge the "List of Orders" and "Order Details" tables into a new single table named "Orders Data" based on the "Order ID" relationship.

Home → Merge Queries → Merge Queries as New

First table: List of Orders
Second table: Order Details
Click Order ID column in both tables
Left Outer (All from first, matching from second)
Click OK

Click the expand icon at the top of that column.
Select Product, Quantity, Sales, Profit, etc.
Click OK

Rename Orders Data
Orders Data

Handling Missing Data & Duplicate Data:

● Identify missing values in the data and determine a strategy to address them.

Select Table→View→Check→Column Quality, Column Distribution, Column Profile→All Column is Valid No Error and Null Values

● **Check for duplicate rows and define a strategy to handle duplicates.**

Home → Keep Rows → Keep Duplicates→empty → so no duplicates

Sorting and Filtering Data:

● In the 'Orders Data' table, utilize sorting and filtering techniques on columns like Order Date, State or Category to analyze data based on specific criteria:

◆ Sort the orders by Order Date in descending order to analyze recent trends.

Select OrderID→Click the Corner→Decening Order

◆ Filter the orders to focus only on a specific state (e.g., Tamil Nadu) for regional analysis.

Select State→Click the Corner→Filer-Tamilnadu

```
= Table.SelectRows(#"Sorted Rows1", each ([State] = "Tamil Nadu"))
```

		Order Date	Customer Name	State	City	Location
1		02/09/2019	Kalyani	Tamil Nadu	Chennai	Chennai Tamil Nadu
2		02/09/2019	Kalyani	Tamil Nadu	Chennai	Chennai Tamil Nadu
3		02/09/2019	Kalyani	Tamil Nadu	Chennai	Chennai Tamil Nadu
4		02/09/2019	Kalyani	Tamil Nadu	Chennai	Chennai Tamil Nadu
5		07/11/2018	Surabhi	Tamil Nadu	Chennai	Chennai Tamil Nadu
6		04/08/2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu
7		04/08/2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu
8		04/08/2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu
9		04/08/2018	Aarushi	Tamil Nadu	Chennai	Chennai Tamil Nadu

Grouping and Aggregating Data:

● Duplicate the "Order Details" table and calculate the count of each Order ID, average profit by Category or total amount by Sub-Category.

Calculate the count of each orderID

Select **Order Details Summary** table.

Go to **Home → Group By**.

Settings:

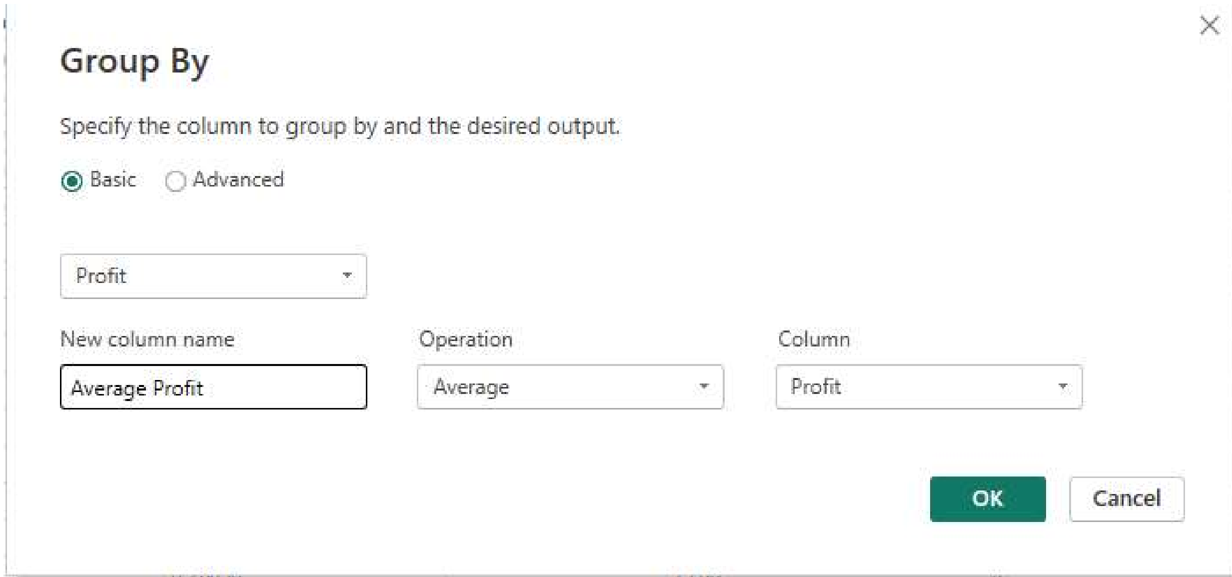
Column Operation

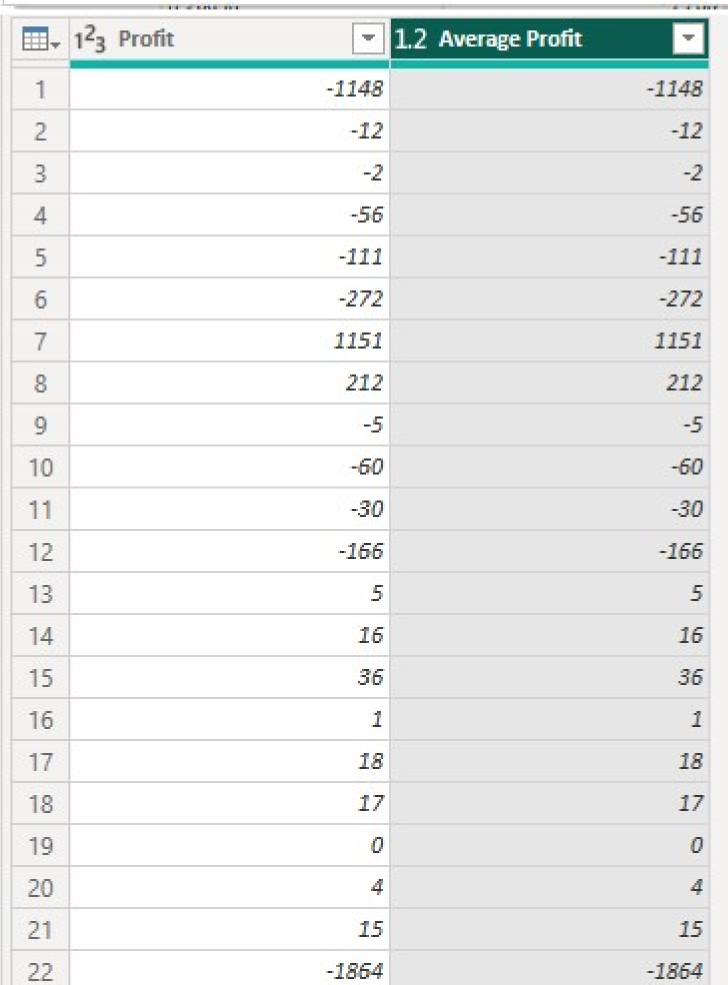
Order ID Count Rows Ok

```
= Table.Group(#"Sorted Rows1", {"Or
```

	Order ID	Count
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2

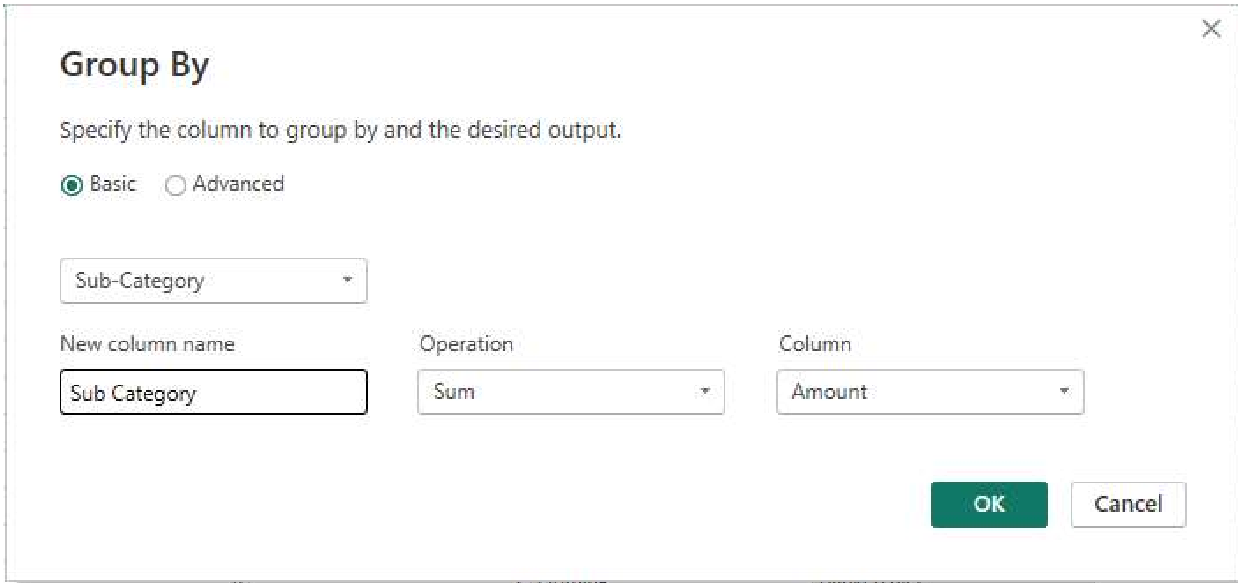
Average Profit by Category





	Profit	Average Profit
1	-1148	-1148
2	-12	-12
3	-2	-2
4	-56	-56
5	-111	-111
6	-272	-272
7	1151	1151
8	212	212
9	-5	-5
10	-60	-60
11	-30	-30
12	-166	-166
13	5	5
14	16	16
15	36	36
16	1	1
17	18	18
18	17	17
19	0	0
20	4	4
21	15	15
22	-1864	-1864

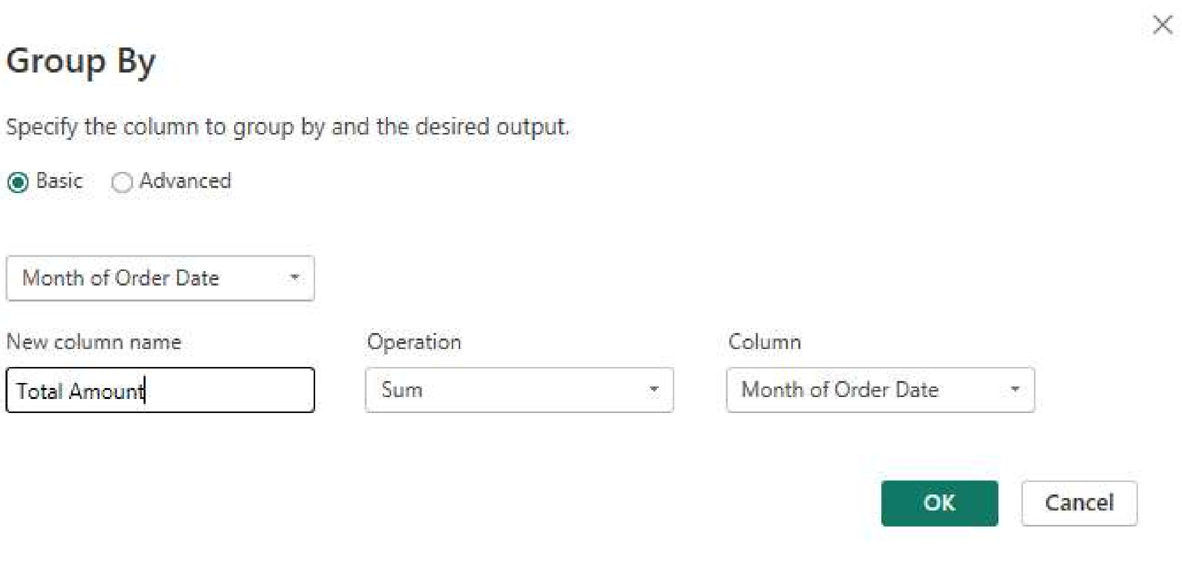
Total Amount by Sub-Category



	Sub-Category	Total Amount
1	Bookcases	23032
2	Stole	7394
3	Hankerchief	7128
4	Electronic Games	12054
5	Phones	14761
6	Saree	26298
7	Trousers	9746
8	Chairs	19031
9	Kurti	1571
10	T-shirt	2903
11	Shirt	3568
12	Leggings	579
13	Tables	16070
14	Printers	26450
15	Accessories	11137
16	Furnishings	6387
17	Skirt	822

● Duplicate the “Sales Target” table and aggregate the total target amount by Month of Order Date.

Sales target→Duplicate→Rename the table as Monthly Target Summary.



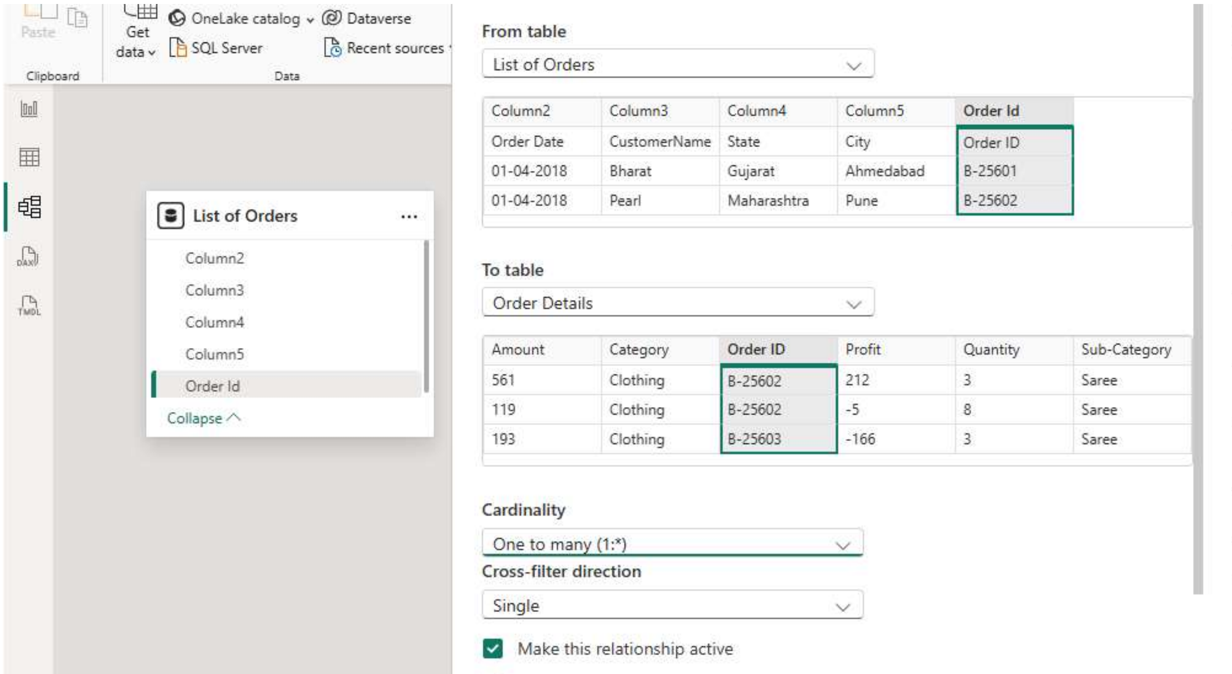
Select month of order date – select sort ascending

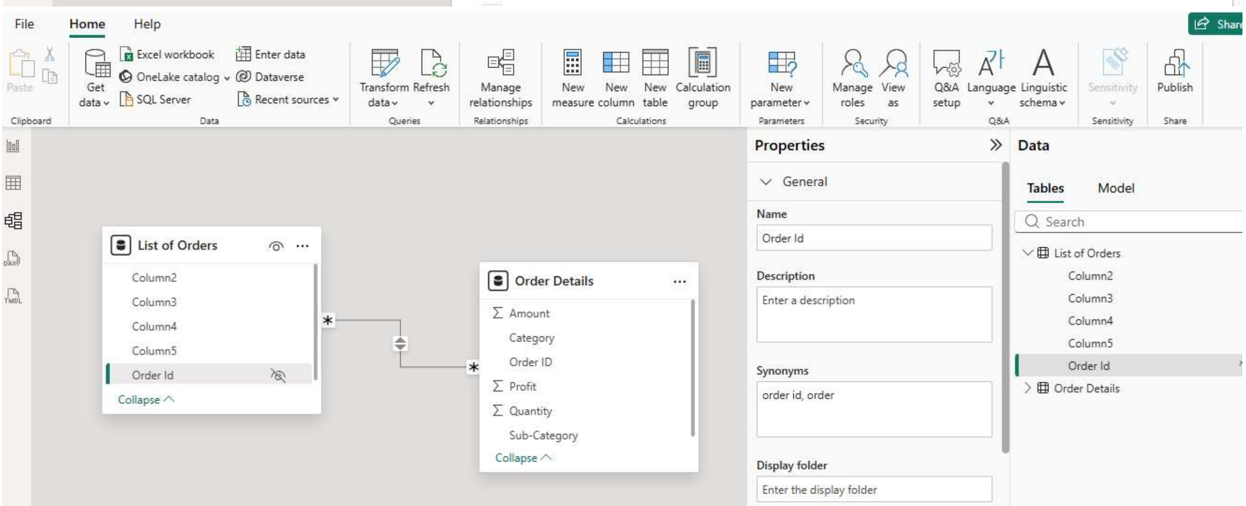
	Month of Order Date	1.2 Total Target
1	19/01/2026	43500
2	19/02/2026	43600
3	19/03/2026	43800
4	18/04/2026	31400
5	18/05/2026	31500
6	18/06/2026	31600
7	18/07/2026	33800
8	18/08/2026	33900
9	18/09/2026	34000
10	18/10/2026	36100
11	18/11/2026	36300
12	18/12/2026	36400

Data Modeling:

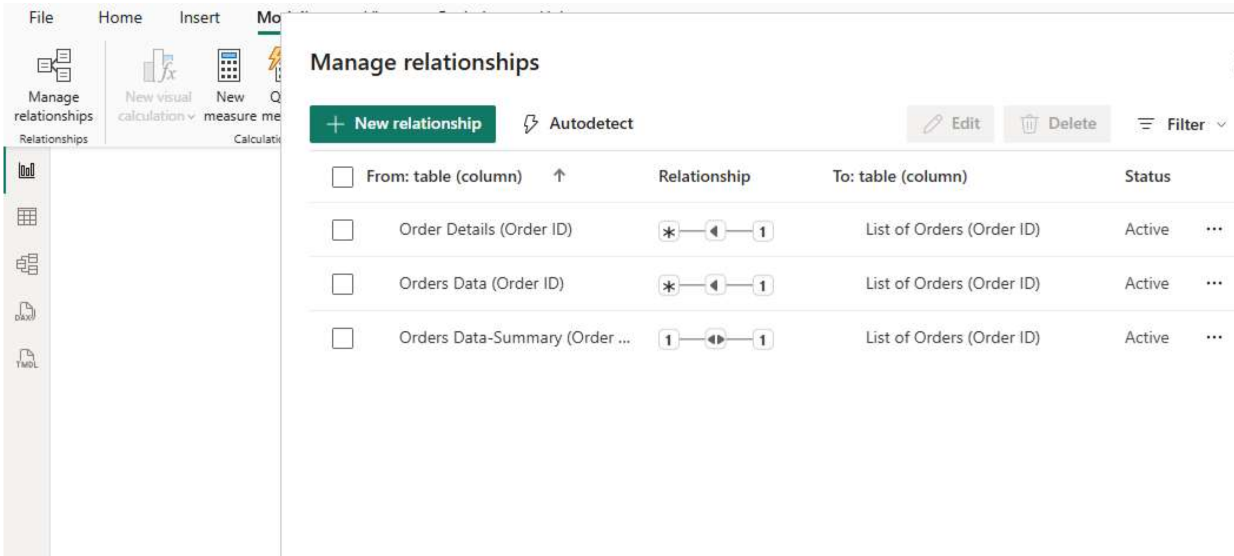
● Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.

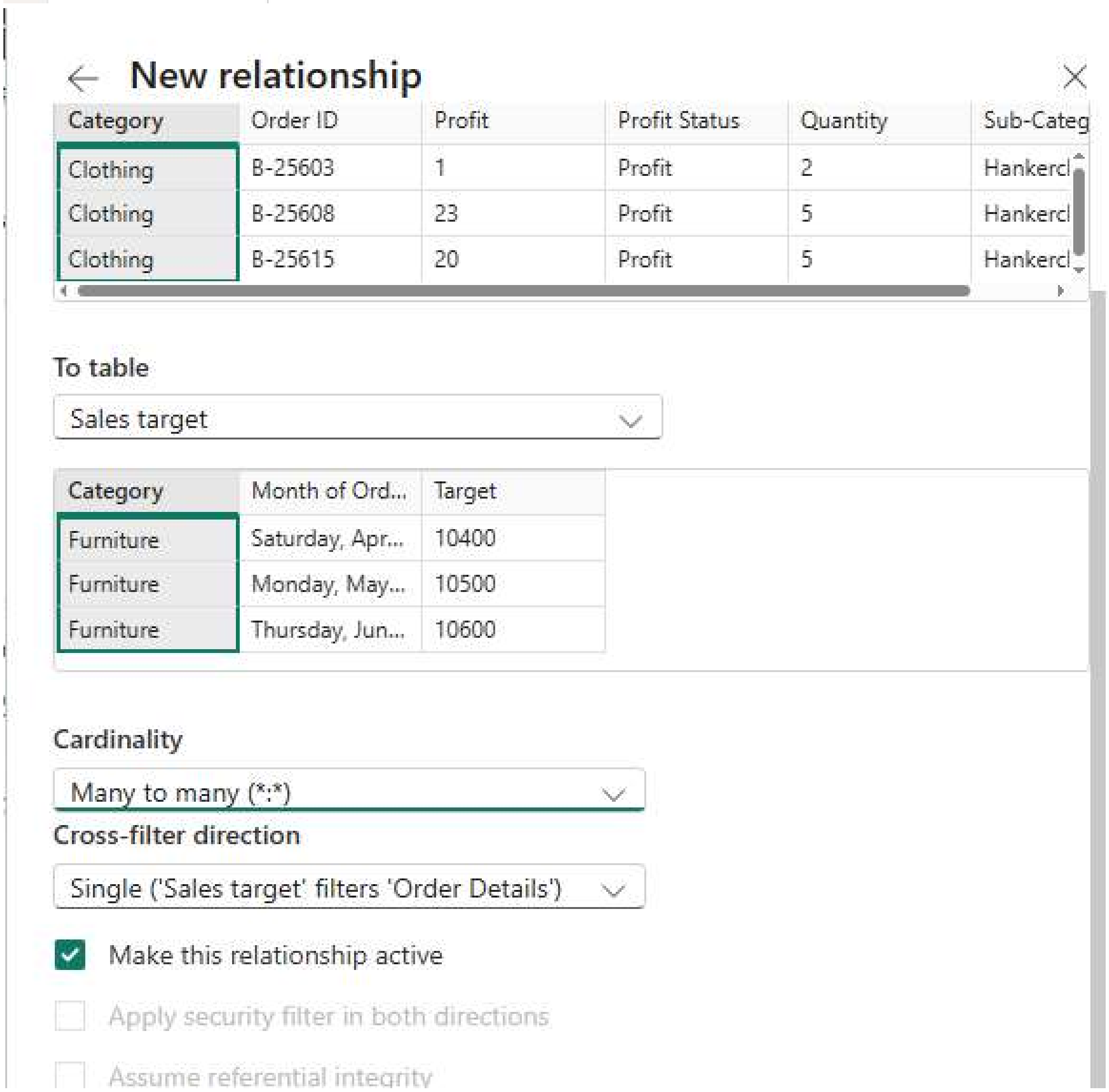
Import “List of Order” and “Order Details” → Click Model view→

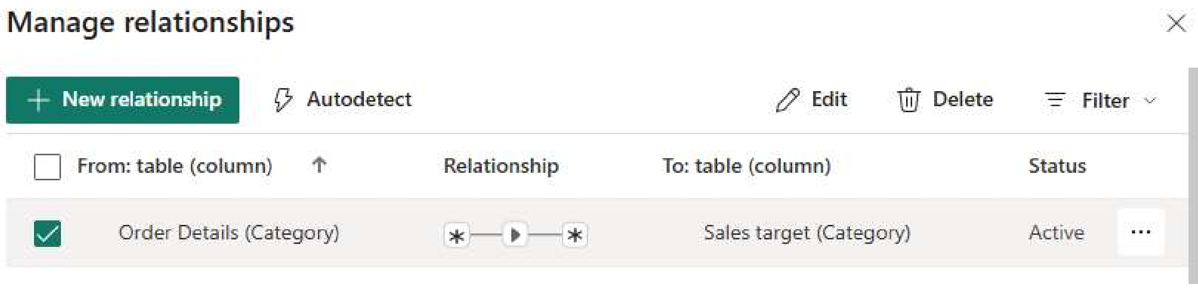

Paste
Get data
OneLake catalog
Dataverse
SQL Server
Recent sources
Clipboard
Data
List of Orders
Column2
Column3
Column4
Column5
Order Id
Collapse
From table
List of Orders
Column2 | Column3 | Column4 | Column5 | Order Id
Order Date | CustomerName | State | City | Order ID
01-04-2018 | Bharat | Gujarat | Ahmedabad | B-25601
01-04-2018 | Pearl | Maharashtra | Pune | B-25602
To table
Order Details
Amount | Category | Order ID | Profit | Quantity | Sub-Category
561 | Clothing | B-25602 | 212 | 3 | Saree
119 | Clothing | B-25602 | -5 | 8 | Saree
193 | Clothing | B-25603 | -166 | 3 | Saree
Cardinality
One to many (1:*)
Cross-filter direction
Single
Make this relationship active


File
Home
Help
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Publish
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
Q&A
Sensitivity
Share
List of Orders
Column2
Column3
Column4
Column5
Order Id
Collapse
Order Details
Amount
Category
Order ID
Profit
Quantity
Sub-Category
Collapse
Properties
General
Name
Order Id
Description
Enter a description
Synonyms
order id, order
Display folder
Enter the display folder
Data
Tables
Model
Search
List of Orders
Column2
Column3
Column4
Column5
Order Id
Order Details

● Build a relationship between the “Order Details” and “Sales Target” tables based on the ‘Category’ column. Click "Manage relationships" and ensure this relationship is active.






Manage relationships
New relationship
Autodetect
Edit
Delete
Filter
From: table (column)
Relationship
To: table (column)
Status
Order Details (Category)
Sales target (Category)
Active